

# Kunj Rathod

AI Researcher — Mechanistic Interpretability & LLM Systems

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## EDUCATION

### B.S. Computer Science — GPA 3.7/4.0, Dean's List

University of Utah

Aug 2023 – Dec 2026

Salt Lake City, UT

- Relevant: Machine Learning, NLP, Interpretability of LLMs, Distributed Systems, Multimodal LLM Agents

## RESEARCH

### Mechanistic Interpretability — Circuit-Level Generalization in LLMs

2026 – Present

University of Utah, advised by Ana Marasovic / w/ Emma Brown, Christina Dong

- Extracted sparse feature circuits from gemma-2-2b using GemmaScope cross-layer transcoders (Wu et al. framework) to predict whether circuits learned on default tasks generalize to counterfactual task variants.
- Built automated attribution graph pipelines using Pathways Discovery (PD); synthesized prior work across 15+ interpretability papers on circuit stability, feature reuse, and generalization correlation.
- **Key finding:** suppressing over-refusal SAE features unexpectedly worsened over-refusal rates rather than reducing them — a publishable negative result with direct implications for activation-steering-based safety interventions.
- Prepared NeurIPS-style paper; research poster presented April 2026.

### Long-term Active Embodied QA (LA-EQA)

Spring 2026

University of Utah / w/ Nikesh Subedi

- Co-authored “Exploring Long-term Active Embodied QA in Simulated Indoor Environments.” Proposed **Video Mind Palace (VMP)**: replaced Robotic Mind Palace’s offline GPT-4o scene-graph pipeline with direct Qwen3-VL video queries, achieving **31–57% inference speedup** at 13–17% relative accuracy drop (100-question benchmark, A100 GPU).

### Spatial Memory for Embodied & Long-Horizon Web Agents

2026 – Present

University of Utah

- Designing retrieval-augmented memory architectures unifying spatial, temporal, episodic, and semantic memory; evaluating on OS Marathon, BrowseComp, Mind2Web benchmarks.

## EXPERIENCE

### Software Engineer Intern — Azure Data

May 2026 – Present

Microsoft

Redmond, WA

- Building scalable distributed data systems on Azure; full-stack engineering within the Azure Data ecosystem.

### AI Services Intern — SUDO Program

Jan 2025 – Present

University of Utah Health

Salt Lake City, UT

- Built HIPAA-compliant LLM platform serving 90+ hospital executives via AWS Bedrock Agents, Knowledge Bases, and Guardrails; achieved p95 TTFT <200ms and 40% latency reduction with DynamoDB+S3 session persistence at 1,000+ concurrent conversations.
- Owned API design, schema, UI (React/TypeScript), and Lambda orchestration across 6 features shipped in 4 sprints.

### AI Engineering Intern — Legal AI

Nov 2024 – Apr 2025

CourtEasy.ai / Nugen

Remote

- Benchmarked 8 LLM families (InLegalBERT, InLegalLLaMA, GPT-4o-mini) on LegalBench, NyayaAnumana, IL-TUR, LegalBench-RAG across F1, latency, cost; analysis cut projected inference spend \$50k+/yr.
- Built hybrid RAG (dense + BM25 + reranking) over 10M+ legal documents; +28% retrieval accuracy, -35% hallucination rate.

### AI Research Intern — BioGraphRAG

May 2024 – Aug 2024

Garje Marathi Global

Salt Lake City, UT

- GraphRAG over 1M+ biomedical entities (UniProt, AlphaFold, RXNav) on NebulaGraph; +40% factual accuracy, 3× graph traversal speedup to sub-500ms p95. Article republished by NebulaGraph.

## SELECTED PROJECTS

### FNDR — Privacy-First Local AI Memory (macOS)

2024 – Present

Rust, Tauri, Metal, ONNX, Llama 3.2, Whisper, Apple Vision, MCP

- Zero-trust local RAG assistant: Metal-accelerated on-device LLM/VLM inference, Apple Vision OCR, CLIP embeddings, Graphiti-style temporal knowledge graph, local Whisper transcription, MCP server for agent interop.

### HirePilot — 9-Agent Autonomous Recruiting System [Podium AI Hackathon — 1st Place]

2026

TypeScript, Anthropic API, Twilio, Google Calendar, Slack

- End-to-end autonomous hiring pipeline: GitHub sourcing, live voice interviews via Twilio + transcript analysis, multi-dimensional scoring, manager approval loop via Slack.

WRITING & PUBLICATIONS

- **Exploring LA-EQA in Simulated Indoor Environments** — Rathod, Subedi. Class report, CS Advanced AI, U of Utah, Spring 2026.
- **BioGraphRAG** — co-authored technical article; republished by NebulaGraph. [Substack](#), Oct 2024.
- **LLM Families on Legal Benchmarks** — internal comparative analysis, CourtEasy.ai / Nugen, 2025.

TECHNICAL SKILLS

|           |                                                                                                                                              |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Languages | Python, C++, Java, TypeScript, Rust, SQL                                                                                                     |
| AI / ML   | PyTorch, Sparse Autoencoders, Attribution Graphs, RAG, GraphRAG, LangChain, LlamaIndex, Transformers, Multi-Agent Systems, Embodied AI, VLMs |
| Infra     | AWS Bedrock, Lambda, S3, DynamoDB, Docker, PostgreSQL, NebulaGraph, ChromaDB, Pinecone                                                       |

AWARDS

**Kahlert Impact Prize** (\$1,000) — Kahlert School of Computing, U of Utah, Mar 2026 · **Dean’s List** 2024–25 · **Minute0 Hackathon Winner** Feb 2026